Multi-Person Human Mesh Recovery

with the Anny Body Model

A smaller student model for recovering 3D people

from a single image

Muhammad Saad Najib

Master’s thesis · Supervisor presentation

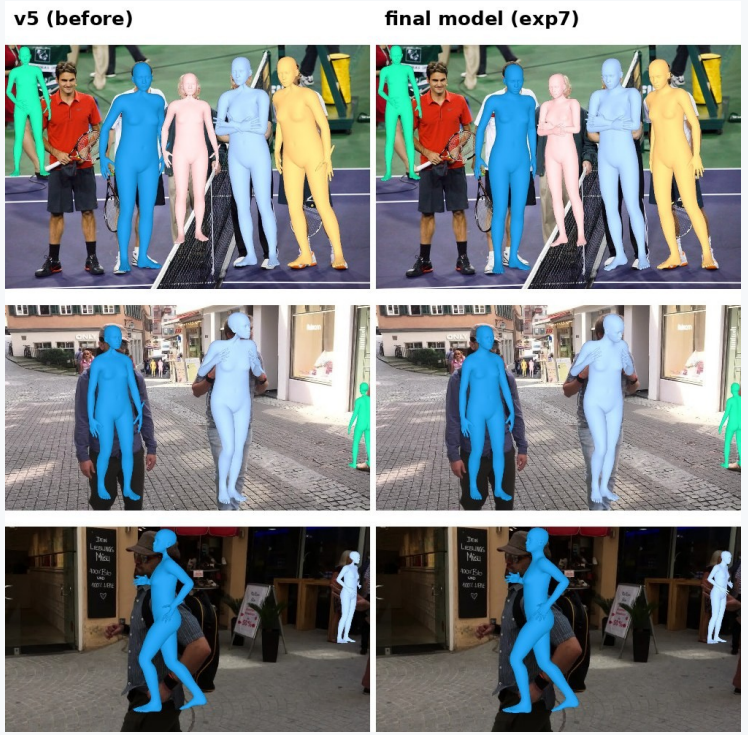


Illustration: Base/v5 and teacher overlays

These are not student predictions.

One photograph, several difficult decisions

Recover a whole body

Estimate pose, body shape and distance from the camera.

People can overlap or be partly hidden.

Real photographs

Synthetic images have exact labels but differ from photos.

3DPW annotations use a different body model.

Reduce inference cost

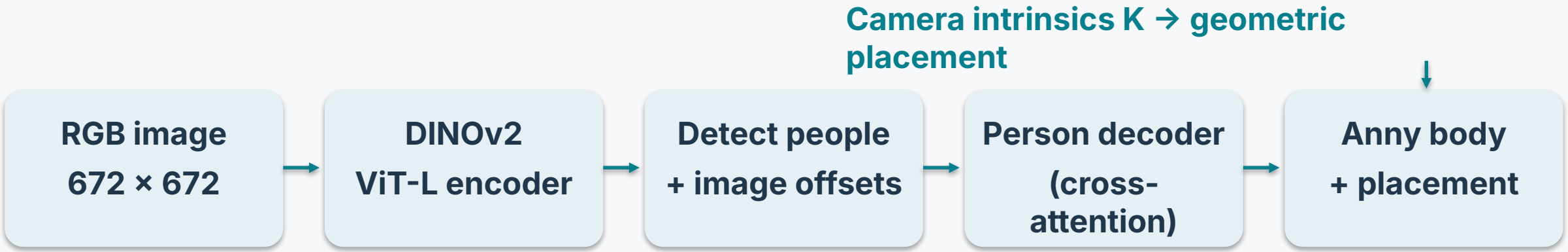
Large models require more computation and memory.

Can a smaller model retain useful reconstruction quality?

Research question: how much accuracy can a smaller student retain?

Baseline: one image → multiple 3D bodies

Baseline used in this project: the released Multi-HMR model with Anny outputs.



Encoder: extracts image information

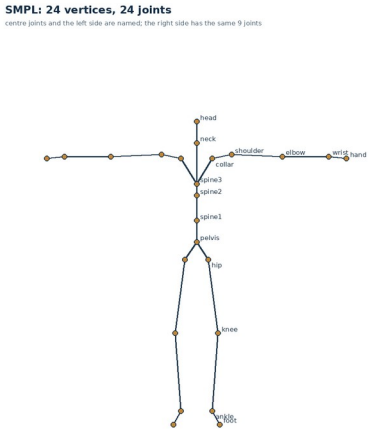
Decoder: predicts one body per detected person

Single-shot: process the whole image in one forward pass.

Sources: Multi-HMR (Baradel et al., 2024); released NAVER Anny checkpoint; thesis §4.2

Two body models: SMPL and Anny

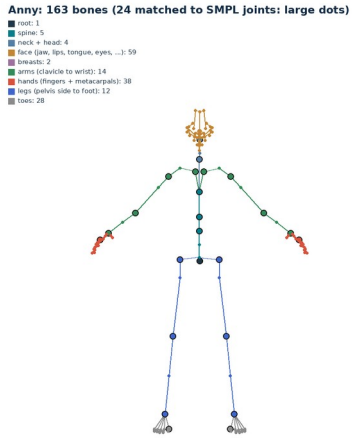
SMPL · the 3DPW labels



24 joints
Centre (6):
pelvis, spine1–3,
neck, head
Each side (9):
hip, knee, ankle,
foot, collar, shoulder,
elbow, wrist, hand

Mesh: 6,890 vertices · 24 joints (23 + root)
Shape: 10 betas: abstract PCA directions from adult scans; no single value has a direct meaning
Pose: 24 × 3 axis-angle rotations (72 numbers)

Anny · what our model predicts



163 bones
- root 1
- spine 5
- neck + head 4
- arms 7 × 2
- hands (fingers) 19 × 2
- legs 6 × 2
- toes 14 × 2
- breasts 2
- face 59

large dots: the 24 points fitted to SMPL

Rig: 163 bones: 75 predicted, 88 helpers kept at rest
Shape: 11 phenotypes, each in [0, 1], human-readable:
gender · age · height · proportions · muscle · weight
cupsize · firmness · african · asian · caucasian
Pose: our model predicts 163 × 6D rotations

Why convert 3DPW? We predict Anny, but real video labels exist only as SMPL, and AnnyOne is synthetic.
So I refit the 3DPW SMPL bodies with Anny: 24 matched points; the root and 18 bones move, 144 of 163 bones stay at rest.

Sources: thesis §2.1, §3.3, App. B; scripts/data/smpl_to_anny.py; model.py; Anny rig: figures/anny_rest_skeleton.json

Two data sources with one common body format

AnnyOne: synthetic images

Exact Anny pose, shape and camera labels.

579,860 training images
100-image thesis holdout

70% of mixed training samples

3DPW: real video frames

Original labels describe SMPL bodies.
Convert them into Anny training targets.

Train 5,882 · Validation 2,622 · Test 8,190
30% of mixed training samples

The 100-image synthetic holdout contains adults only; it cannot establish child performance.

Sources: thesis Chapter 3; numbers_data.tex

How the 3DPW–Anny labels were created

1

Read source labels

3DPW joints, shape, camera and frame ID

2

Fit an Anny body

Optimize root, pose, then free shape values

3

Share body shape

One median shape for each person's sequence

4

Build image labels

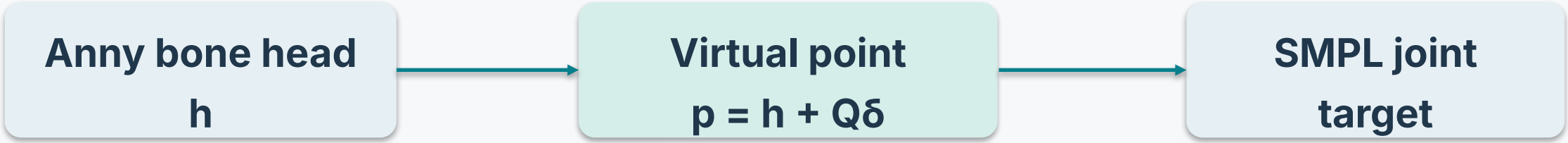
Refit poses; transform to camera coordinates

Fit every third valid frame on a shared frame grid; validate and filter the completed labels.

Sources: thesis §3.3–3.5; smpl_to_anny.py, converter v2.3

Virtual joints compare equivalent body locations

A geometric offset changes the point used for fitting, while the Anny skeleton stays intact.



Move by a fixed offset δ, rotated with the bone by Q

Minimize the gap to the target

Joint definitions can be aligned; limb lengths and body proportions can still differ.

Sources: thesis §3.4 and Appendix B

Converted labels: checked, with residual warnings

Original image | Source SMPL annotation | Converted Anny annotation



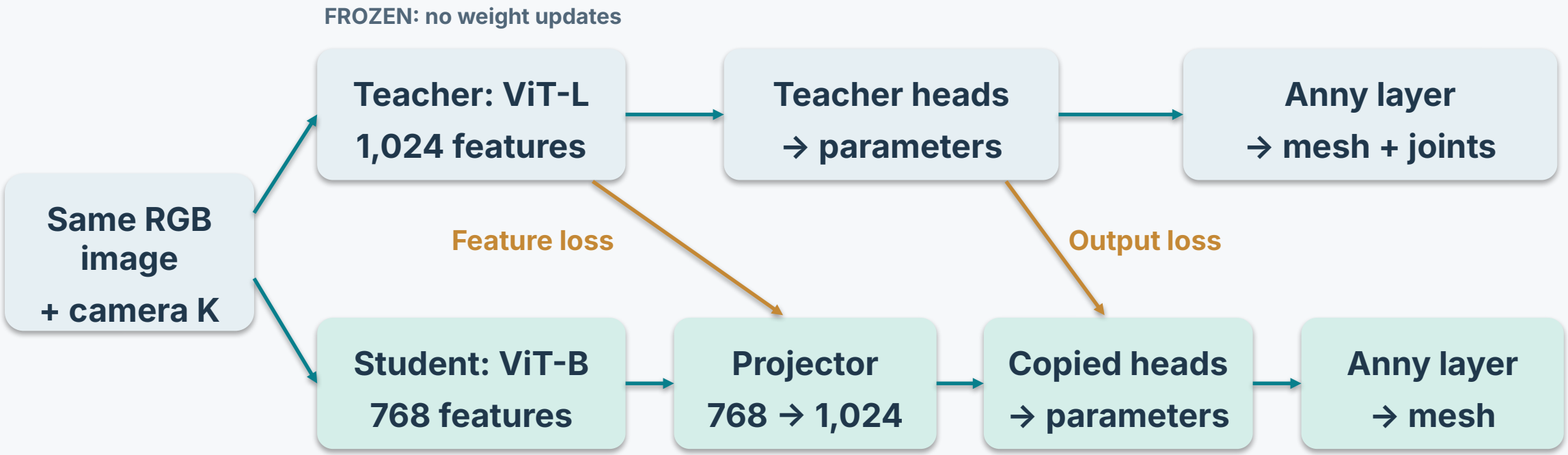
14.1 mm median body-joint discrepancy

WARN: remaining limb-proportion and joint-definition differences.

This label-fit discrepancy is not the model’s prediction error.

Sources: thesis §3.5, Table T05; F14_label_check_downtown_walking_00771.jpg

The student learns from labels and a frozen teacher



Dataset labels → supervised losses on student predictions

Deployment uses only the student branch, including its projector.

Sources: thesis §4.1, §4.8; model.py; distill.py; train.py

What goes in, and what comes out?

A body model turns predicted parameters into a visible 3D surface.

Input

RGB image + camera K

Training also supplies per-person body labels and anchor locations.

Predictions per person

Detection score + 2D offset
163 bone rotations
11 body-shape parameters
Head-anchor depth

Decoded output

3D body mesh + joints

Pose: how the body bends
Shape: body proportions
Placement: where it stands

Sources: thesis §4.2; model.py

Supervised losses: compare predictions with labels

Loss group	What is compared?	Plain-language purpose
Detection + offset	Person-anchor score and 2D location	Find each person correctly
Pose + shape	Bone rotations and body parameters	Recover articulation and proportions
Depth + translation	Head depth and 3D anchor position	Put the body at the right distance
3D joints + vertices	Pelvis-centred predicted / target bodies	Match the body geometry
2D projection	Predicted / target points in the image	Make the mesh line up with the photo

$L_{total} = L_{supervised} + L_{feature\ KD} + L_{output\ KD}$

Sources: thesis §4.3; Table T02; exact formulas in backup slides

Knowledge distillation: two kinds of teacher guidance

Feature distillation

Compare teacher image features with the student's projected features.

KL divergence, temperature T = 4.
"Learn similar image representations."

Output distillation

Compare corresponding people's rotations, shape parameters and depth.

Weighted L1 differences.
"Learn similar body predictions."

The student learns from the true labels and the teacher; both KD weights are 1.0.

Sources: thesis §4.8; distill.py; loss.py

Training recipe for the reported student

30
epochs

1,000
steps per epoch

8
images per batch

70 / 30
synthetic / real mix

Initialize ViT-B + copy teacher heads → **First 5 epochs keep copied heads frozen** → **Continue training, then calibrate outputs**

Learning rate: 3×10^{-5}; halved every 12 epochs. Teacher stays frozen.

70/30 is the sampling ratio of images, not people. Test images are excluded.

Sources: thesis Table T11; exp8_distill_vitb.sh; job 3614191

How reconstruction quality is measured

Errors are measured on matched people. Lower error is better; higher recall is better.

Metric	Simple meaning	Unit
PVE	Average surface-point error after pelvis alignment	mm ↓
PA-PVE	Surface error after optimal rotation, translation and scale alignment	mm ↓
MPJPE	Average joint-position error after pelvis alignment	mm ↓
Recall	Share of labelled people recovered by the detector	% ↑
Latency / model cost	Time per forward pass; parameters and operations	ms / M / GFLOPs ↓

Sources: thesis §4.10; Table T07

The student loses accuracy relative to its teacher

PVE / PA-PVE, in mm. All 3DPW values use converted Anny labels.

Model	Synthetic holdout	3DPW test	3DPW recall
Released baseline	60.3 / 44.6*	126.9 / 68.8	97%
Teacher (Final, ViT-L)	73.7 / 50.3	89.0 / 57.6	99%
Student (ViT-B)	119.6 / 80.8	110.1 / 71.8	99%

* Baseline synthetic images are potentially in its training set.

Baseline real-image training exposure also differs from teacher/student.

Student: lower 3DPW PVE than the released baseline, but higher error than the teacher.

Sources: thesis Tables T16–T17; jobs 3614191, 3611336, 3614192

The student is smaller and faster

Same benchmark: 672-pixel input, one person, batch 1, fp32 with TF32.

Model	Parameters	Compute	Median forward time
Teacher	318.38 M	1,927.4 GFLOPs	59.7 ms
Student	101.38 M	603.3 GFLOPs	28.5 ms

3.14×
fewer parameters

3.19×
less computation

2.10×
faster forward pass

This is a model-forward benchmark, not end-to-end video FPS.

Source: thesis Table T16; paired L40S measurements, job 3614191

Synthetic example: inspect placement as well as pose

Historical comparison: Base/v5 versus the larger teacher (Final).

Input image | Dataset ground truth | Base / v5 | Teacher / Final



Meshes use each model’s own predicted depth. These panels are not student results.

Sources: thesis Figure F12; AnnyOne holdout image 579860, V1

Real-photo examples show strengths and limits

Image | Base/v5 overlay | Image | Teacher/Final overlay



Legacy demos: box-based placement and rest-pose fingers; not raw depth or hand-accuracy evidence.

Sources: thesis Figures F10; walking and garden examples; demo rendering settings

Video comparison: student and released baseline

Completed run: man_walking.mp4 · 356 frames · NVIDIA L40S

Same 356 frames · Student: 356/356 matched · Baseline: 354/356 matched

INSERT YOUR COMPARISON VIDEO HERE

Input | Student | Released baseline

Inspect jitter, sudden pose changes and changing body shape.

Frame coverage does not prove smoothness. Video FPS has not yet been verified.

Sources: video-compare-3642415.out; thesis video evidence; demo_video.py

What remains unresolved

Accuracy gap

Student errors remain higher than the teacher's.

The 95% accuracy-retention target is not met.

Real-image body shape

Children and body-shape variation remain difficult.

The adult-only holdout cannot test child performance.

Temporal evidence

Review the actual video.

Measure synchronized FPS and frame-to-frame stability.

Sources: thesis Discussion and Conclusion; video evidence

A smaller student with a measured trade-off

01 Converted real 3DPW annotations into checked Anny labels.

02 Trained a ViT-B student with label and teacher supervision.

03 Measured a 2.10× speedup, with higher reconstruction error.

Main result: a clear speed–accuracy trade-off, plus a working video-comparison pipeline.

Backup: supervised objective and exact weights

$$L_{sup} = 10L_{det} + L_{off} + 0.1L_{rot} + 20L_{shape}$$

$$+5L_{depth} + 5L_{trans} + 100(L_{j3D} + L_{v3D}) + 0.1(L_{j2D} + L_{v2D})$$

$$L_{rot} = \frac{1}{N_h}\sum_{i}\sum_{b} m_{ib} w_b \|\hat{R}_{ib} - R_{ib}\|_1$$

L1 = sum of absolute differences; rotations use matrix entries.

N_h = supervised people; m = availability mask; w = normalized bone weight.

3DPW hand rotations are masked; mesh losses still contain rest-pose hands.

Depth and translation refer to the same head anchor used for placement.

The final recipe supervises all shape entries, including converter constants.

Sources: thesis §4.3; Table T02; loss.py

Backup: distillation equations

$$p^t = \mathrm{softmax}(Z^t/T), \quad p^s = \mathrm{softmax}(\tilde{Z}^s/T), \quad T = 4$$

$$L_{feat} = \frac{T^2}{B} \sum_{u,n,c} p^t_{unc} \log \frac{p^t_{unc}}{p^s_{unc}}$$

$$L_{out} = \frac{1}{N_h} \sum_i [0.1 \sum_b \| R^s_{ib} - R^t_{ib} \|_1 + 20 \sum_k |\phi^s_{ik} - \phi^t_{ik}| + 5 |d^s_i - d^t_i|]$$

B: batch size · n: image token · c: feature channel · i: person · b: bone · k: shape value

Feature KL averages over the batch and sums tokens/channels.

Output KD uses matched labelled-person locations and detached teacher predictions.

Rotation KD does not use the supervised hand mask or per-bone boost.

Sources: thesis §4.8; distill.py; loss.py rotation_distill_term

Selected references and evidence

Method background

Baradel et al. (2024). Multi-HMR: Multi-Person Whole-Body Human Mesh Recovery in a Single Shot.

Brégier et al. Human Mesh Modeling for Anny Body. arXiv:2511.03589.

Oquab et al. (2023). DINOv2: Learning Robust Visual Features without Supervision.

von Marcard et al. (2018). Recovering Accurate 3D Human Pose in the Wild Using IMUs and a Moving Camera.

Hinton et al. (2015). Distilling the Knowledge in a Neural Network.

Project evidence

Revised thesis: commit 3e2f21e · Student results: job 3614191

Video execution: job 3642415 · Every pictured model is identified on its slide.